

Application: are unlearned capabilities still in the weights?

The elicit-vs-teach metrics on WMDP-unlearned Zephyr-7B

Question. The main results show that the metrics separate a capability that is already in the weights (fine-tuning *elicits* it) from one that is absent (fine-tuning *teaches* it). Unlearning claims to move a model from the first case to the second. If it did, relearning is teaching and every metric should read “teach”; if the capability is only hidden, relearning is elicitation and the metrics should read “elicit”.

Setup in six lines. The original model is Zephyr-7B-beta. Four public unlearned versions of it are tested: RMU, ELM, NPO, SimNPO. The capability is WMDP-bio, 4-option multiple choice, scored as the argmax over the four answer letters (chance 0.25). The items are split at random into halves A and B. *Relearning* fine-tunes on A as text facts (question and correct answer, no options; LoRA $r=64$) and is read on the held-out half B, which it never sees. The *fine-tuning null* is the same recipe on MMLU facts, which cannot teach bio answers. There is no never-learned model, so the elicit reference is the *original* model (before training) and the *original’s own relearning* (after training); the teach reference is each metric’s own null, and the elicit/teach values of the main results are given alongside.

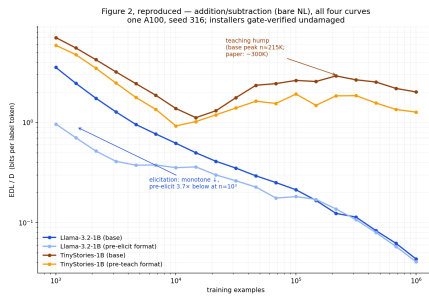
Reading a cell. Each cell is the measured value and the call it makes. After relearning (Tables 1 and 3): **elicit** (reads like the original: at least half of the original’s distance from the null), *partly* (above its own null, less than half), **teach** (at its own null). Before training (Table 2) the only question is whether the capability can be read in the unlearned model: **readable**, *partly*, or not read. That reading is one-sided: what a metric reads is there, but what it cannot read may still be there, hidden before or after the point the metric looks at (Section 4). — means there is nothing to measure (the model has no task behaviour for this metric). A metric is called only where this design measures both its null and the original’s value; the others are shown next to the original’s value without a call (the appendix table).

1. Does it come back, and what brings it back?

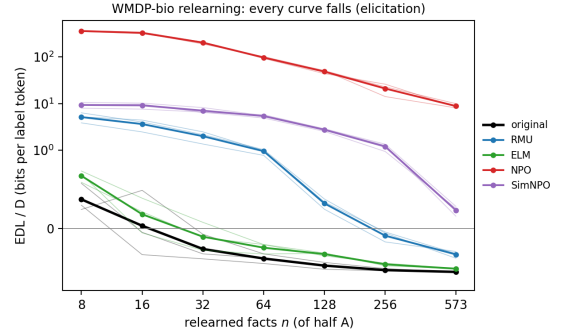
- **Every method leaves the capability in the weights:** relearning half the items brings the other half back to the original’s accuracy, and costs what the original’s own relearning costs.
- **What unlocks it differs:** for NPO and SimNPO any small fine-tune does; for RMU only bio-specific training does; ELM is in between.
- **The original paper’s own signature agrees** (M19, Figure 1): the information relearning extracts per label token falls with the number of facts for every method, in every seed and under both floors, as it does for the original. Teaching would show a rising phase, as the TinyStories arm does in panel (a). What an unlearned model needs beyond the original is paid on the first facts and shrinks with every doubling of n .

	original	RMU	ELM	NPO	SimNPO
accuracy on B: the model as given	0.67	0.31	0.35	0.26	0.42
after an unrelated (MMLU) fine-tune	—	0.35	0.55	0.62	0.62
after relearning on A	—	0.68	0.67	0.64	0.66
<i>the same with no options shown: the correct answer picked among the four, given the question alone (null 0.31)</i>					
as given	0.44	0.39	0.42	0.32	0.32
after an unrelated (MMLU) fine-tune	—	0.39	0.44	0.43	0.45
after relearning on A	—	0.47	0.48	0.46	0.46
M17 relearning beats the unrelated fine-tune	—	0.68 vs 0.35 elicit	0.67 vs 0.55 elicit	no room —	no room —
M17u the unrelated fine-tune unlocks B	—	locked: 0.31→0.35	0.35→0.55 elicit	0.26→0.62 elicit	0.42→0.62 elicit
M18 relearning cost over the original’s (nats)	0	+0.15 ± 0.19 elicit	+0.03 ± 0.14 elicit	+4.6 ± 9.0 —	−0.31 ± 0.53 elicit
M19 EDL/D falls with n (bits, $n=8 \rightarrow 573$)	0.37 → −0.56	5.1 → −0.34 elicit	0.67 → −0.52 elicit	355 → 8.8 elicit	9.3 → 0.23 elicit
MMLU (general ability)	0.58	0.58	0.59	0.47	0.48

Table 1: Accuracy on the held-out half B. M17: position between the unrelated-fine-tune value and the original, 90% interval RMU 1.04 [0.88, 1.19], ELM 0.98 [0.57, 1.40]; for NPO and SimNPO the unrelated fine-tune already reaches the original, so there is no room left to measure. M17u: position between the unlearned model and the original, RMU 0.11 [−0.02, 0.23], ELM 0.64 [0.50, 0.78], NPO 0.89 [0.78, 1.00], SimNPO 0.80 [0.62, 0.99]. M18: prequential code length of relearning (each batch scored before training on it) minus the original’s, net of the same difference on the unrelated fine-tune; 0 = as cheap as the original relearning facts it knows; $\pm$ is 90%; NPO’s first steps repair a broken output, so its interval is too wide to read. M19: the original paper’s signature (Donoway et al., §4), EDL per label token of relearning on n facts of A, mean of three seeds, from $n = 8$ to $n = 573$; falling with $n =$ elicitation, a rising phase = teaching (Figure 1). The no-options rows score the correct answer’s first token against the other three answers’, none shown (the relearning format); the null reassigned the “correct” candidate at random. The picture is the same there: every relearned model reaches the original, RMU beyond its unrelated fine-tune (0.47 vs 0.39), the others already there after it; ELM is largely there before any training.



(a) reference: the project’s Figure-2 replication



(b) WMDP-bio relearning

Figure 1: The EDL sweep. (a) The project’s replication of the paper’s Figure 2 (arithmetic; decisions.md 2026-08-22): the model that has the capability (Llama-3.2-1B, blue) falls monotonically; the model that must be taught it (TinyStories-1B, brown) falls, then rises $2.6\times$ (the teaching hump), then saturates. (b) Relearning on the first n facts of a seeded order of A, bits per label token, floor = the relearned model’s own validation loss (OCV; the paper’s test floor, the loss on B, gives the same calls); thick = mean of three seeds, thin = each seed; linear between -1 and 1 , logarithmic beyond. Every curve falls, the original’s included; no method has a rising phase. The excess over the original per label token shrinks from $n = 8$ to 573 by $21\times$ (RMU, $4.8 \rightarrow 0.22$ bits), $7\times$ (ELM, $0.30 \rightarrow 0.04$), $38\times$ (NPO, $355 \rightarrow 9.4$) and $11\times$ (SimNPO, $9.0 \rightarrow 0.79$): a fixed cost paid on the first facts, not a cost per fact. NPO’s level is its output repair (its first batch costs about 240 nats per label token). The original ends below zero because its validation floor is harder than its training facts (Caveats).

2. Before any training: is the capability still readable in the unlearned model?

	metric	main results: elicit / teach	original	RMU	ELM	NPO	SimNPO
M12	hidden preference, bio (nats)	2.31 / 0.06	3.33	0.84 <i>partly</i>	0.40 <i>partly</i>	0.42 <i>partly</i>	3.52 readable
M12	hidden preference, cyber (nats)	2.31 / 0.06	3.57	0.34 <i>partly</i>	0.13 <i>partly</i>	2.32 readable	3.31 readable
M16'	answer probe, bio (accuracy; shuffled ≈ 0.28)	—	0.60	0.28 not read	0.28 not read	0.47 readable	0.49 readable
M16'	answer probe, cyber	—	0.46	0.28 not read	0.29 not read	0.41 readable	0.39 readable
M9	answer depth: lens, best middle layer (nats)	2.1 / 0.0	4.88	1.18 <i>partly</i>	0.39 <i>partly</i>	0.17 <i>partly</i>	0.45 <i>partly</i>
M11	repeatable task circuit (split-half J@32)	0.60 / noise	0.64	noise not read	noise not read	noise not read	0.46 readable
M14	answer carried by the state (swap gain)	+0.39 / +0.05	+0.66	+0.08 not read	+0.09 not read	+0.03 not read	+0.38 readable
M15	option-reading heads (heads; flip rate)	22 / 0 heads	15; 0.73	20; 0.52 readable	0 not read	0 not read	57; 0.70 readable
M12f	correct answer preferred among the four with no options shown (top-1; null 0.31)	—	0.43	0.35 not read	0.40 readable	0.29 not read	0.34 not read
M11f	repeatable circuit on the no-options surface (split-half J@32)	—	0.49	0.42 not read	0.60 readable	0.28 not read	0.23 not read
M1★	original's 32 heads destroy U's answer (random 0.05); machinery, Section 4	—	0.93	0.99 [†]	no answer —	no answer —	0.89
M10★	original's head states patched into U (accuracy)	—	0.49	0.20→0.30 <i>partly</i>	0.27→0.48 readable	0.00→0.00 not read	0.13→0.25 <i>partly</i>
M2★	wiring, U vs original (J@256; ceiling 0.55)	0.36 / —	—	no answer —	no answer —	no answer —	0.23 <i>partly</i>
M13	accelerating descent (negative share; blocks 20, 28)	1.00 / 0.00	0.00	0.00 —	0.00 —	1.00 —	0.00 —
tally (readable / partly / not read)				1 / 4 / 6	3 / 3 / 5	3 / 2 / 6	7 / 3 / 2

Table 2: Values on the unlearned model itself, no training. “Main results” gives the elicit parent and the teach parent from the arithmetic study where the metric has one. M16' adapts M16: the probe reads the correct option (4 classes), not the operands. M14: how often swapping the state at the answer position makes the model prefer the other item's answer, minus the rate with no swap. M10★: accuracy of U before and after the original's states at its 32 top heads are written in (32 random heads: RMU 0.20, ELM 0.34, SimNPO 0.16). [†] RMU's remaining answer signal is small (logit difference +0.75 vs +4.87 in the original); the original's heads carry all of it, but its MMLU-circuit heads carry it equally (Section 4), so the row is not tallied. M12f and M11f are read with no options in the prompt (question, then “Answer:”): M12f scores the correct answer's first token against the other three answers' first tokens, the null reassigns the “correct” candidate at random ($p < 0.01$ counts as readable; RMU $p = 0.05$, SimNPO $p = 0.06$); M11f is M11 on that surface. M13 cannot be called here: the original is at a minimum of its own bio loss (negative share 0.00, unlike the main results' latent elicit parent at 1.00), so the metric has no elicit reference; NPO alone sits on a strongly concave slope ($\lambda_{\min} = -1.5 \times 10^5$), where gradient ascent left it, and any descent step speeds up. Curvature is restricted to decoder blocks 20 and 28 (the full Hessian does not fit a 7B model on one GPU). M8 (latent reach) has no WMDP analogue and is not run.

- **SimNPO** reads like the original on 7 of 12: on the multiple-choice prompt the circuit, the state, the heads and the probe are all intact, and the output is pushed away from the answer. With no options shown the answer is not read either.
- **NPO**: the answer is readable in the state (probe 0.47 vs original 0.60) while the output says nothing, on either surface, so the damage is at the read-out.
- **ELM**: nothing is readable on the multiple-choice prompt, but with no options shown ELM still prefers the correct answer (0.40; original 0.43, null 0.31), and its circuit on that surface is repeatable: the suppression is bound to the format. Writing the original’s head states in turns the multiple-choice answer on (0.27 $\rightarrow$ 0.48; the original scores 0.49 on this test).
- **RMU** hides it best: the probe, the circuit and the state swap cannot read it, and with no options shown it is at the edge of the null (0.35, $p = 0.05$). Yet relearning brings it back in full on both surfaces (Table 1), so what these metrics miss in RMU is present, not absent.

3. After relearning on A: does the relearning look like eliciting or teaching?

Every relearned child is compared with the *original’s* relearned child, trained with the same recipe on the same facts the original already knows; that child is the elicit reference in this setting.

- **After relearning, all four read elicit on almost every called metric.** On the no-options surface the child’s circuit overlaps the original’s at 0.39–0.60, above the split-half ceilings (0.31–0.37) and well above the cross-subject overlap (0.21–0.28); on the multiple-choice prompt the child uses the original’s answering machinery (overlap 0.64–0.73), which any four-option question uses (Section 4).
- The exception is patching a child’s states into the unlearned model (M10), which fails where the unlearned model’s own output is broken (NPO, SimNPO) or scrambled (RMU).

	metric	main results: elicit / teach	orig. re- learned	RMU	ELM	NPO	SimNPO
M1f	circuit overlap with the original on the no-options surface, heads (J@32; chance 0.02; cross-subject 0.21–0.28)	0.33 / 0.02	ceiling 0.31–0.37	0.60 elicit	0.46 elicit	0.46 elicit	0.39 elicit
M17f	held-out B with no options shown, beyond the unrelated fine-tune	–	0.44	0.47 vs 0.39 elicit	no room —	no room —	no room —
M10	child’s head states patched into U (accuracy)	switches on / –	–	0.21→0.31 teach	0.33→0.54 elicit	0.00→0.00 teach	0.14→0.20 teach
	tally , with Table 1 (elicit / partly / teach)			5 / 0 / 1	5 / 0 / 0	2 / 0 / 1	3 / 0 / 1
<i>multiple-choice circuit rows: the same values against the original’s MMLU circuit (Section 4), so not called:</i>							
M1	circuit overlap with the original, heads (J@32; vs its MMLU circuit)	0.33 / 0.02	ceiling 0.60; own 0.56	0.73; 0.60	0.64; 0.56	0.68; 0.52	0.68; 0.52
M1	original’s 32 heads destroy the child’s answer (bio heads; MMLU heads)	0.73 / –	0.86; 0.84	0.88; 0.85	0.86; 0.84	0.86; 0.83	0.83; 0.84
M2	wiring overlap with the original (J@256; vs its MMLU wiring)	0.36 / –	ceiling 0.27; own 0.27	0.32; 0.32	0.32; 0.34	0.33; 0.32	0.29; 0.31
M4	circuit at step 1 vs final (J@32; ceiling)	at step 1 / mid-run	–	0.45 (0.78)	0.73 (0.73)	0.49 (0.73)	0.42 (0.73)
<i>reported, not called (no null or teach value is measured for these here):</i>							
M3	head roles shared with the original (J; chance 0.01)	0.69 / 0.88	–	0.13	0.08	0.17	0.15
M5	gradient fades (first 1% / last 10%)	3.4× / 0.7×	3.2×	14.5×	6.1×	134×	13.7×
M6	weight write $\ \Delta W\ /\ W\ $ ($\times 10^{-3}$)	50 / 86	8.8	8.9	8.8	11.2	11.0
M7	state change: one shared push (PC1)	0.06 / 0.57	0.64	0.49	0.57	0.69	0.66

Table 3: Values on the relearned child, read on B. “Original relearned” is the elicit reference: the same training on the original, which knows the facts. M1f: the child’s circuit for answering with no options shown against the original’s circuit on the same surface; the ceilings are the split-half overlaps of the circuits themselves, and “cross-subject” is the same children against the original’s MMLU circuit on that surface (the original’s own bio circuit against its MMLU circuit there: 0.21). M17f: the no-options accuracy of Table 1; “no room” where the unrelated fine-tune already reaches the original. The multiple-choice circuit rows give, after the semicolon, the same measurement against the original’s MMLU circuit (or with the original’s MMLU-circuit heads ablated); the two are equal within 0.1 and at the split-half ceiling, so these rows show the original’s answering machinery in use, not bio knowledge, and are not tallied. M4: overlap of the circuit after the first training step with the final one, in brackets the ceiling. M10: accuracy of U with the child’s states at the child’s 32 top heads written in; the child itself scores 0.54–0.57. M3, M5, M6 and M7 are shown next to the original’s relearning but not called, because the main results’ scale does not carry over and this design has no teach value to replace it: the original’s own relearning reads 0.64 on M7, past the main results’ teach value; LoRA and the same stopping rule fix the size of M6’s write; M5’s first steps are inflated by repairing the unlearned output; and M3 did not separate elicit from teach in the main results either (0.69 vs 0.88). The tally counts M17, M18 and M19 from Table 1 (M17u is not counted: it asks what unlocks the capability, not whether it is there).

4. Why the calls are mixed, and how each metric is judged

Before training, a call can only be one-sided. Unlearning is meant to stop the capability from showing, so a metric that looks where the suppression acts, or after it, finds nothing, while one that looks before it, or goes around it, still finds the capability. Mixed calls therefore say where each method hides the capability; they do not contradict each other. Whether it is still in the weights is answered by the tests that bring it back (M17, M17u, M18, M19, M10★), and those read elicit for all four.

- **Hidden at the read-out: NPO and SimNPO.** The answer is readable in the state (probe 0.47 and 0.49; original 0.60, shuffled labels 0.28; through the layers where the original’s answer is readable, 0.63 and 0.67 of the original’s excess) while the output map does not carry it (lens 0.04 and 0.12 of the original’s; accuracy on B 0.26 and 0.42; original 0.67). Both train the output probability of the forget-set answers down, and their weight change is spread over 29 of 32 layers and the output head. NPO was trained on bio only, and its cyber answers still show at the output (hidden preference 2.32; original 3.57).
- **Hidden before the answer forms: RMU and ELM.** Even the probe cannot read the answer at any layer (0.28, the shuffled level), so the suppression acts inside the network before the state carries the answer. The weights say where: 90% of RMU’s change sits in the MLPs of layers 5–7, and 90% of ELM’s in layers 4–7 (its adapter). Going around that edit brings the capability back: writing the original’s head states into ELM restores its accuracy to 0.48 (original 0.49), and ELM still answers with no options shown (0.40; original 0.43), so its suppression is bound to the multiple-choice format. Relearning restores both, and it does not repair the edit: every child’s LoRA write puts the same share into the edited layers as the original’s own relearning does (RMU 10%, ELM 13%).

Machinery or knowledge? A multiple-choice reading needs the machinery that reads the options and writes the letter as well as the knowledge, so every metric read on that prompt was run again on the MMLU split (Table 4). The machinery is intact in all four models: on MMLU, the hidden preference, the probe, the state swap and the option-reading heads read at 0.57–1.22 of the original’s excess; NPO and SimNPO show partial read-out damage there (lens 0.43 and 0.24; their MMLU accuracy is 10 points below the original’s). So the bio drops on the probe, the state swap and the heads are knowledge-specific: RMU and ELM keep the MMLU reading at the original’s level and lose the bio reading entirely. The circuit metrics are the exception. A relearned child’s circuit overlaps the original’s MMLU circuit as much as its bio circuit (heads 0.52–0.60 vs 0.64–0.73, both at the split-half ceiling; wiring equal), and the original’s MMLU-circuit heads are as necessary as its bio-circuit heads, in the unlearned models and in every child. Those rows show the original’s answering machinery in use and are no longer tallied. On the no-options surface the circuits are bio-specific (0.39–0.60 against a cross-subject 0.21–0.28), and that reading (M1f) is tallied instead.

Caveats

- There is no teach anchor from this model: without a model that never learned WMDP-bio, “elicit” means “like the original”, and the teach values come from each metric’s own null and from the main results.
- The metrics before training are read on all bio items, those after relearning on half B; the first 512 items in source order score lower (original 0.51) than B (0.67), so the two blocks are not compared with each other.
- The multiple-choice circuit metrics (M1, M2, M4, the necessity of the original’s heads, M1★) read the shared answering machinery, as their MMLU controls show (Section 4), and are not tallied; the tallied circuit reading is the no-options one (M1f). The probe, the state swap and the option heads

	metric	RMU	ELM	NPO	SimNPO
<i>before training: position r on bio on MMLU (control); 1 = the original, 0 = the null</i>					
M12	hidden preference	0.25 0.99	0.12 0.91	0.13 0.89	1.06 1.22
M16'	answer probe	0 1.02	0 1.05	0.59 0.90	0.67 0.98
M9	lens, middle layers	0.24 0.99	0.08 0.89	0.04 0.43	0.09 0.24
M14	answer carried by the state	0 0.96	0 0.89	0 0.64	0.57 0.57
M15	option-reading heads	0.70 0.91	0 0.98	0 0.70	0.96 0.82
M1★	original's heads necessary in U: bio heads MMLU heads	1.06 1.11	–	–	0.89 0.94
M10★	original's states patched in: at bio heads at MMLU heads	0.33 0.32	0.92 0.83	0 0	0.28 0
<i>after relearning: overlap with the original's bio circuit with its MMLU circuit (the original's own: 0.56, 0.27, 0.21)</i>					
M1	heads, multiple-choice prompt (J@32)	0.73 0.60	0.64 0.56	0.68 0.52	0.68 0.52
M2	wiring, multiple-choice prompt (J@256)	0.32 0.32	0.32 0.34	0.33 0.32	0.29 0.31
M1f	heads, no options shown (J@32)	0.60 0.26	0.46 0.26	0.46 0.28	0.39 0.28
M1	original's heads necessary in the child: bio heads MMLU heads	1.02 1.01	0.99 0.99	0.99 0.98	0.96 0.99

Table 4: The multiple-choice controls. Before training, each metric’s position between its own null (0) and the original (1) on the bio items and, after the bar, on MMLU items, where the knowledge is not in question and only the machinery is measured. A method that keeps the MMLU value and loses the bio value has lost knowledge, not machinery (RMU, ELM on the probe, the state swap and the heads); NPO and SimNPO lose part of the read-out on MMLU too (M9). After relearning, a circuit overlap that is the same against the original’s MMLU circuit as against its bio circuit measures the shared machinery (M1, M2, and the necessity of the original’s heads); the no-options circuit (M1f) is the one that separates bio from MMLU.

have MMLU controls only. The lens has no signal at the middle layers on the no-options surface even for the original (its first-token accuracy there is 0.12), so that read is dropped.

- The no-options surface is a harder read than the multiple-choice one (the original picks the correct answer among the four at 0.44 on B, against 0.67 with the options shown), so its recovery test has less room: for ELM, NPO and SimNPO the unrelated fine-tune already reaches the original there.
- One seed (three for M19) and one relearning recipe; the checkpoint with the lowest validation loss is used because held-out accuracy moves by ± 0.1 between evaluations.
- M19 covers $n \leq 573$, all of half A. In Figure 1a the teaching hump comes after an initial fall and at far larger n , so a hump beyond 573 facts is not excluded by the sweep alone; but the teach arm there ends far above the elicit arm (2.0 vs 0.04 bits at $n = 10^6$), while here the unlearned curves converge on the original’s.
- M19’s OCV floor (the owner’s default) sits on the harder validation rows (the first 10% of A in source order), which is why the original’s curve ends below zero; the paper’s test floor (loss on B) lifts every curve by 0.37–0.41 bits per label token at $n = 573$ (the original ends at -0.15 instead of -0.56) and changes no call. One of the original’s three seeds rises from $n = 8$ to 16.

Appendix: how each metric is judged

	metric	null, measured on the same model	elicit reference	called
<i>before training</i>				
M12	hidden preference	the options permuted	original	yes
M16'	answer probe	probe on shuffled labels	original	yes
M9	lens, middle layers	zero logit difference ($\pm$ SE)	original	yes
M11	repeatable task circuit	type-matched chance	original	yes
M14	answer carried by the state	rate with no swap	original	yes
M15	option-reading heads	no heads found	original	yes
M12f	answer preferred, no options shown	the “correct” candidate reassigned	original	yes
M11f	repeatable circuit, no options shown	type-matched chance	original	yes
M1★	original’s heads necessary in U	100 random head sets	original on itself	machinery
M10★	original’s head states patched in	32 random heads	original (the donor)	yes
M2★	wiring, U vs original	chance	split-half ceiling	yes
M13	accelerating descent	none	none: the original sits at a minimum	no
M7★	state change, original to U	the parent’s own states	describes the unlearning	no
<i>after relearning</i>				
M1f	circuit overlap, no options shown	type-matched chance	split-half ceiling	yes
M17f	held-out recovery, no options shown	the fine-tuning null	original	yes
M1, M2, M4	circuit overlap with the original	type-matched chance	split-half ceiling	machinery
M1	original’s heads necessary in the child	100 random head sets	original’s relearned child	machinery
M10	child’s head states patched into U	32 random heads	the child (the donor)	yes
M17	held-out recovery	the fine-tuning null	original	yes
M17u	an unrelated fine-tune unlocks B	U itself	original	yes
M18	relearning cost	the fine-tuning null	original’s relearning	one-sided
M19	EDL sweep	the paper’s shape rule	original’s own curve	yes
M3	head roles	chance	none measured here	no
M5	gradient fades	none	original’s relearning, 3.2×10^{-3}	no
M6	weight write	none	original’s relearning, 8.8×10^{-3}	no
M7	one shared push	none	original’s relearning, 0.64	no

Table 5: What a single value is compared with. A metric is called only when its null and the original’s value are both measured here; the rest are reported beside the original’s value, since their thresholds come from the main results (full fine-tuning of a 1B model), whose scale does not carry over to LoRA on a 7B model. “Machinery”: the MMLU control reads the same as the bio reading (Table 4), so the row is graded but not tallied. Every before-training metric above M13 also has its MMLU control; the probe, the state swap and the option heads have no no-options version, because their labels (option index, letter swap) need the options.